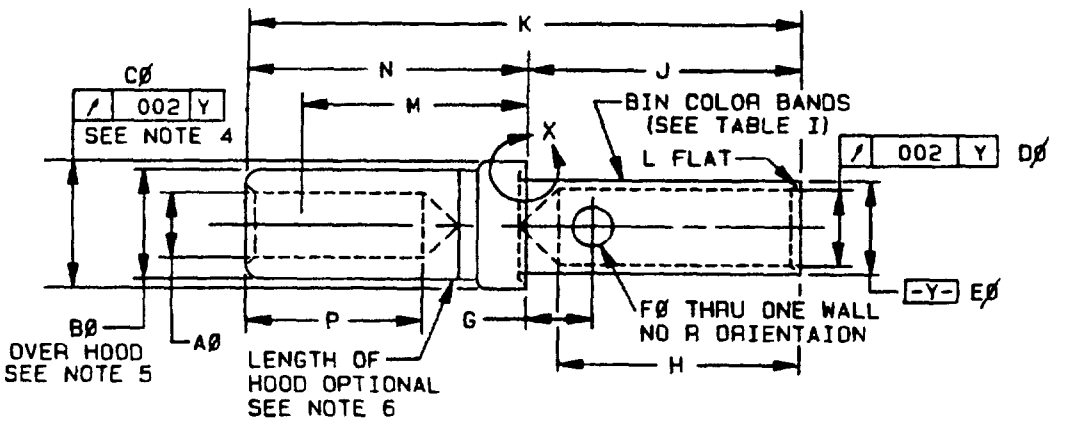
MIL-C-39029/101A
10 March 1989
SUPERSEDING
MIL-C-39029/101
10 December 1985

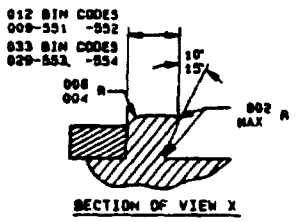
MILITARY SPECIFICATION SHEET

Ⓐ CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE (FOR MIL-S-12883/44, MIL-S-12883/45, MIL-S-12883/46, AND MIL-R-6106/23 RELAY)

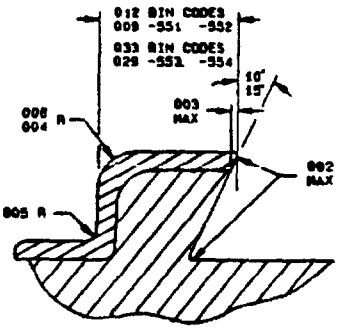
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39029.



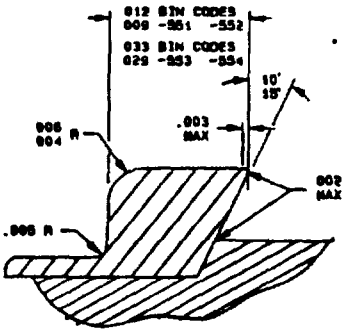


SECTION OF VIEW X



OPTIONAL DESIGN

SECTION OF VIEW X



OPTIONAL DESIGN

SECTION OF VIEW X

Ⓐ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Distance between rear of contact shoulder and point at which a gauge pin, of same basic diameter as mating pin contact and a square face, first engages socket contact spring member.
5. Ⓐ Ⓐ is the closed entry of socket hood.
6. Maximum gap of .010 inch (0.25 mm) between hood and body of the contact.

Inches	mm	Inches	mm	Inches	mm
.002	0.05	.0390	0.991	.0835	2.121
.003	0.08	.0400	1.016	.0855	2.172
.004	0.10	.0410	1.041	.091	2.31
.005	0.13	.042	1.07	.094	2.39
.006	0.15	.0420	1.066	.101	2.57
.009	0.23	.044	1.12	.103	2.62
.012	0.30	.046	1.17	.108	2.74
.018	0.46	.0465	1.181	.120	3.05
.0180	0.457	.048	1.22	.127	3.23
.0200	0.508	.0480	1.219	.130	3.30
.022	0.56	.570	1.448	.137	3.48
.023	0.58	.0590	1.499	.139	3.53
.027	0.69	.060	1.52	.145	3.68
.029	0.74	.0600	1.524	.150	3.81
.0290	0.737	.0610	1.549	.158	4.01
.031	0.79	.0615	1.562	.160	4.06
.0310	0.787	.0630	1.600	.166	4.22
.0320	0.813	.064	1.63	.170	4.32
.033	0.84	.066	1.68	.175	4.44
.0335	0.851	.068	1.73	.180	4.57
.0340	0.864	.070	1.78	.188	4.78
.0355	0.902	.076	1.93	.327	8.31
				.349	8.86

Ⓐ Dimensions.

BIN code	A dia	B max dia	C dia	D dia	E dia	F dia	G	H	J	K ref	L flat	M min	N	P
551	.033	.060	.0615	.0200	.0480	.022	.025	.175	.170	.327	.016	.130	.166	.137
	.031		.0600	.0180	.0465	.018	.021	.160	.160		.0125		.158	.120
552	.033	.060	.0615	.0355	.0480	.022	.025	.175	.170	.327	.008	.130	.166	.137
	.031		.0600	.0335	.0465	.018	.021	.160	.160		.0045		.158	.120
553	.044	.076	.094	.048	.070	.027	.025	.175	.170	.349	.013	.150	.188	.137
	.042		.091	.046	.068	.023	.021	.160	.160		.009		.180	.120
554	.066	.108	.130	.068	.103	.027	.025	.175	.170	.349	.0195	.150	.188	.137
	.064		.127	.066	.101	.023	.021	.160	.160		.0155		.180	.120

FIGURE 1. Socket contact - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.

REQUIREMENTS.

Qualification

(A) The (socket) contact shall be tested on a fully configured relay socket with the applicable MIL-R-6106 relay used as mating plug or pin. (See MIL-S-12883/44, MIL-S-12883/45, MIL-S-12883/46, or MIL-R-6106/23 for applicable mating relay).

Design, construction, and physical dimensions: See figure 1 and table I.

TABLE I. Design characteristics.

BIN code	Color bands			Mating end size (ref)	Wire barrel size	Type	Class
	1st	2nd	3rd				
551	Green	Green	Brown	22	28	A	A
552	Green	Green	Red	22	22		
553	Green	Green	Orange	20	20		
554	Green	Green	Yellow	16	16		

Electrical performance:

(A) Mating electrical contact Relay pin (see applicable MIL-R-6106 specification sheet).

Environmental performance.

Temperature range. -70°C to +125°C

Mechanical performance.

Vibration (sinusoidal): In accordance with MIL-STD-202 method 204, condition G, with the following exceptions:

a The frequency range shall be varied logarithmically between the limits of 10 and 3,000 Hz.

b. The procedure of MIL-STD-202, method 201, may be applied during 10-55 Hz band of the vibration frequency.

(A) Vibration (random). In accordance with MIL-STD-202, method 214, $0.4g^2$ Hz, test condition IG, duration 15 minutes each plane.

Mechanical shock. Perform as specified in MIL-STD-202, method 213, condition C, except peak value shall be 200 g's.

(A) Insertion and withdrawal force: The insertion and withdrawal forces shall be as specified in table IX of MIL-C-39029, except for M39029/101-553 and -554 (see table II). Gauge pin diameters shall conform to the dimensions in table II in lieu of those specified in MS3197

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.

TABLE II. Engagement and separation.

BIN code	Minimum separation force for minimum pin diameter (ounces)	Average engagement force for maximum pin diameter (ounces)	Maximum separation force for minimum pin diameter (ounces)	Minimum separation force for minimum pin diameter (ounces)	Average engagement force for maximum pin diameter (ounces)	Maximum engagement force for maximum pin diameter (ounces)
553	0.7	20	25	0.6	22	29
554	2.0	32	40	1.5	37	46

Probe damage. Testing will be in accordance with MIL-C-39029 with the exceptions listed in table III.

Ⓐ TABLE III. Probe damage.

BIN code	Test probe diameter			Bending moment in-lb ±10% 1/ 2/
	Maximum pin diameter insertion probe (inches) +.0001 -.0000	Minimum pin diameter retention probe (inches) +.0000 -.0001	Moment probe (inches) ±.0001	
551	.0305	.0295	.030	.25
552	.0305	.0295	.030	.25
553	.0410	.0390	.040	1.0
554	.0635	.0615	.0625	2.0

1/ Depth of insertion shall be full depth of bore.

2/ Probe damage test shall be performed in accordance with 4.7.8 of MIL-C-39029, except a collect type holding fixture shall be used to conduct this test. The collect device shall not support the contact, except between the end of crimp barrel where the wire enters and the end of the crimp barrel which comes in contact with the contact shoulder. The contact shoulder and hood shall not be supported.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.

Ⓐ Tools: Associated contact tools shall be as specified in table IV.

TABLE IV. Tools.

BIN code	Basic crimping tool	Positioner	Installing and removal tool
551	M22520/7-01 M22520/1-01	M22520/7-11 K330-3 1/	M81969/16-04 M81969/16-04
552	M22520/7-01 M22520/1-01	M22520/7-11 K330-3 1/	M81969/16-04 M81969/16-04
Ⓐ 553	M22520/7-01 M22520/1-01	M22520/7-12 TH343 1/	M81969/16-01 M81969/16-01
Ⓐ 554	M22520/7-01	M22520/7-13 TH343 1/	M81969/16-02 M81969/16-02

1/ Daniels Manufacturing Corporation part number is specified. Tools may be provided by other vendors if they are determined to be equivalent by the qualifying activity.

Part or Identifying Number (PIN). See table V.

TABLE V Part or Identifying Number (PIN).

BIN code	PIN
551	M39029/101-551
552	M39029/101-552
553	M39029/101-553
554	M39029/101-554

QPL evaluation activity: Defense Electronics Supply Center (DESC-E), Dayton, Ohio 45444-5000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR
Navy - EC, SH, MC
Air Force - 11, 15, 17, 99
DLA - ES

User activities:
Army - AT, ME, AV
Navy - OS

Preparing activity.
Air Force - 85

Agent:
DLA - ES

(Project 5935-3680)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.